Employee Attrition Prediction

B.Tech Mini Project – Project Summary

Objective: build a supervised classification model that estimates whether an employee may leave a company (Attrition = 1) or stay (Attrition = 0), using five HR attributes.

Dataset and selected features

The project uses 1,470 records from the IBM HR Analytics Employee Attrition dataset. The selected input features are OverTime, JobSatisfaction, MonthlyIncome, YearsAtCompany, and WorkLifeBalance. Attrition is the target column.

Methodology

The data was checked for missing values, duplicates, allowed category values, and valid numeric ranges. Overtime and Attrition were encoded as binary values. The dataset was split into 80% training and 20% testing data using stratification. Logistic Regression and Random Forest were compared using 5-fold stratified cross-validation on the training portion. The model with the higher mean F1-score was selected and evaluated on the held-out test set.

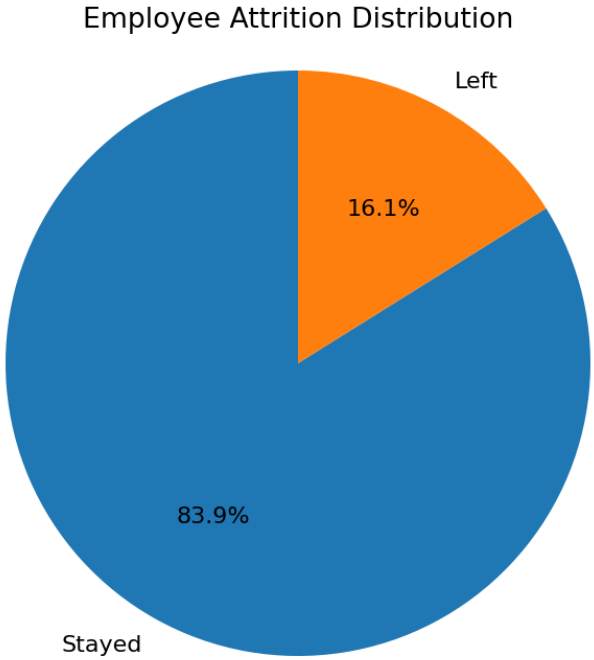
Model evaluation

Metric	Result
Selected model	Random Forest
Test accuracy	79.3%
Attrition precision	39.4%
Attrition recall	55.3%
Attrition F1-score	46.0%

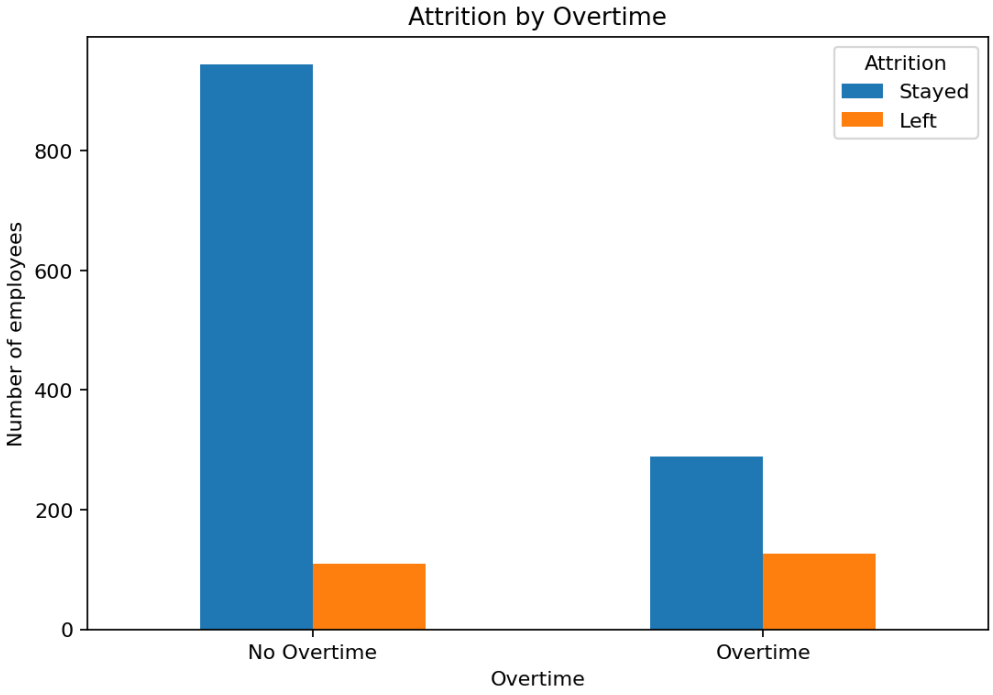
Interpretation: the test accuracy should not be read alone because the dataset is imbalanced. Recall measures how many employees who actually left were identified; precision measures how many predicted leavers actually left. These results are educational and should not be used as the sole basis for employment decisions.

Exploratory Data Analysis Charts

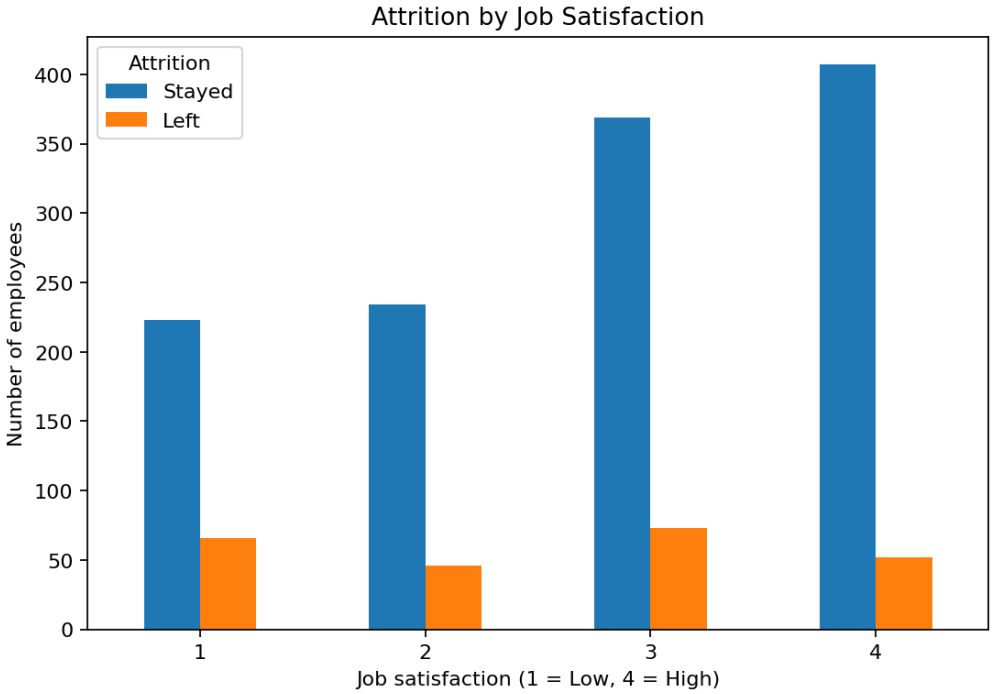
1. Employee attrition distribution



2. Attrition by overtime



3. Attrition by job satisfaction



4. Attrition by years at company

